

AD7190

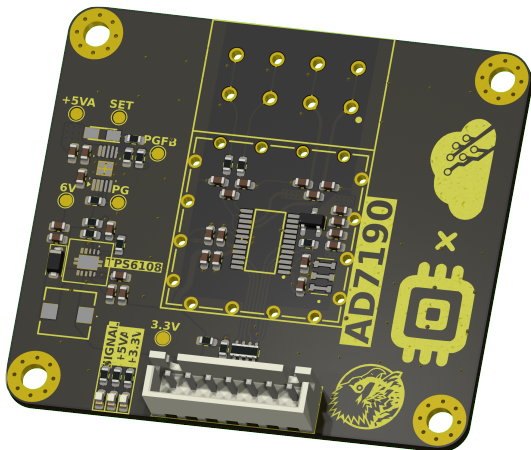
Variant: PRELIMINARY

2026-03-18

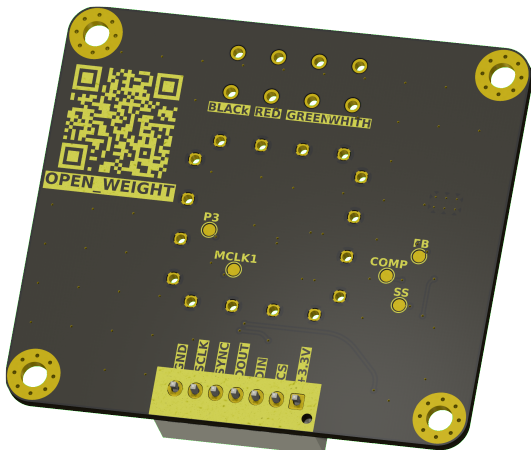
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 18-Mar-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

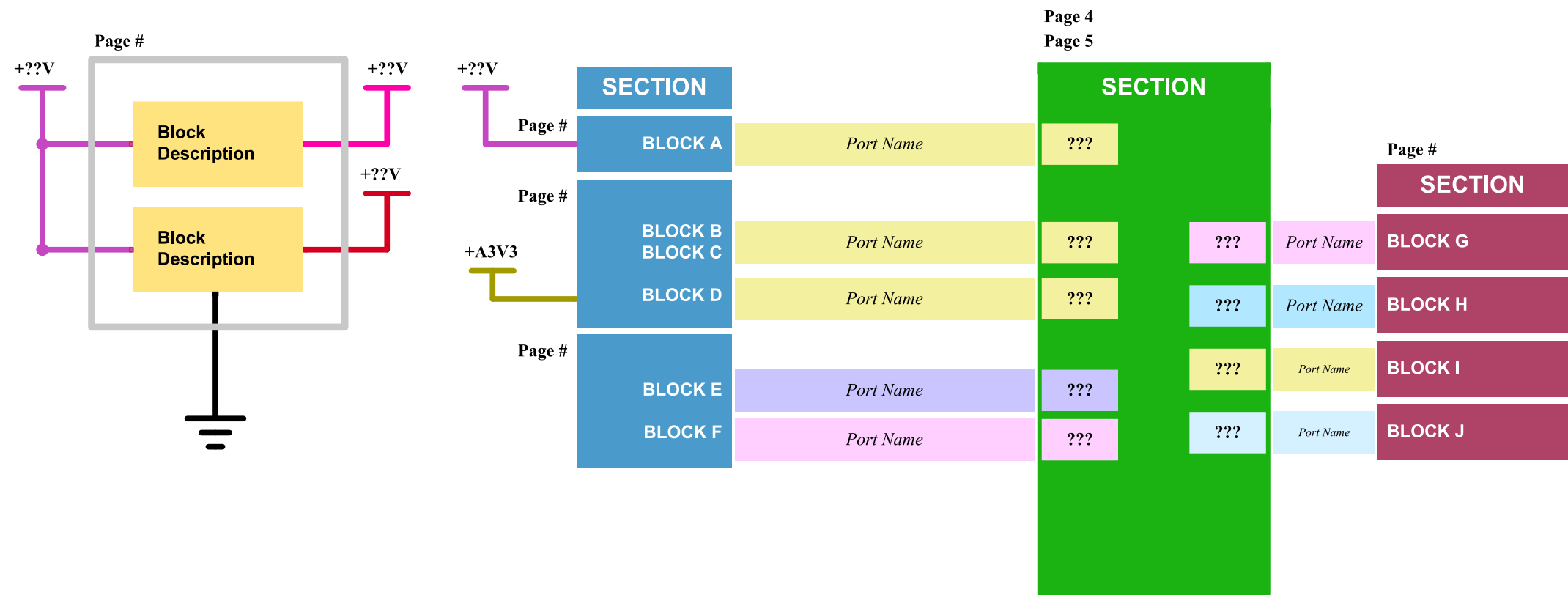
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

	Comments:	Company: OPEN_TRUST_LAB		Variant: PRELIMINARY	Git Hash: 9529cc3
	Sheet Title:	Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Path: /	File Name: Cellule_de_force_V2.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2025-01-12	Revision: + (Unreleased)
			Reviewer:	Size: A3	Sheet: 1 of 10

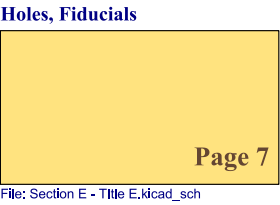
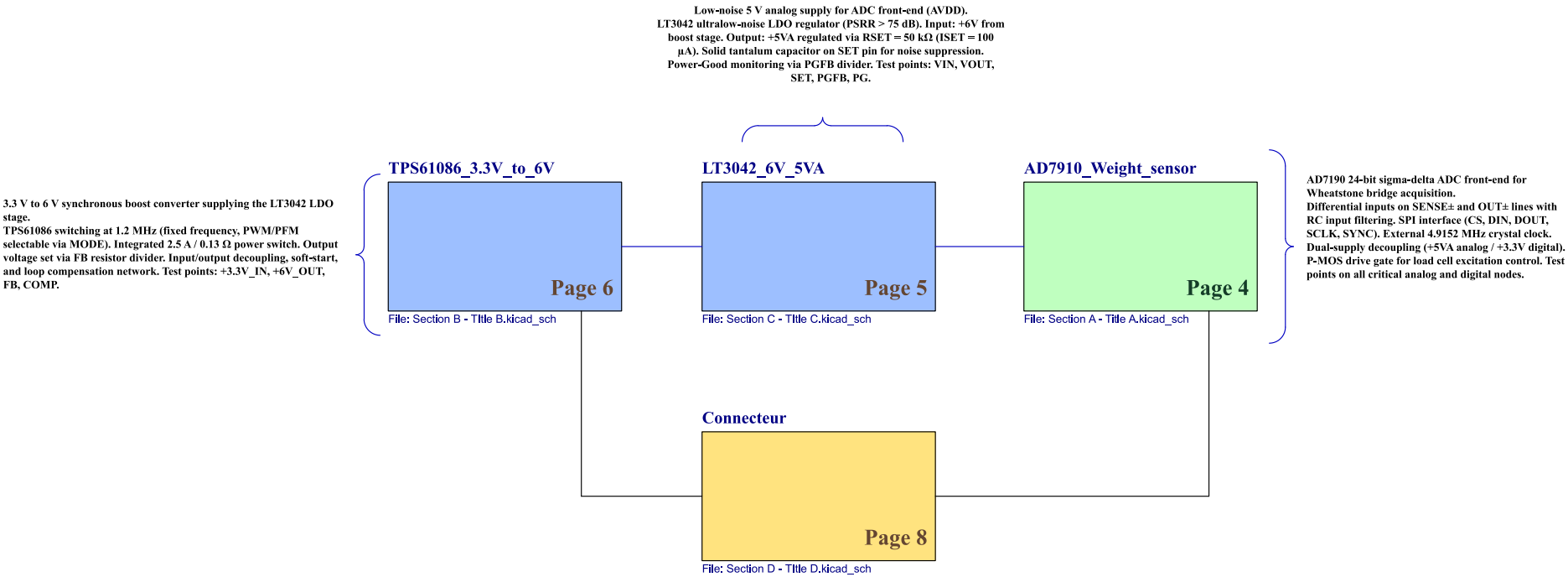
[2] Block Diagram



Target specifications:	
Input voltage:	+3.3 V
Input Amperage	~46-51 mA
Spec 3	??
Spec 4	??

	Comments:		Company: OPEN_TRUST_LAB		Variant: PRELIMINARY	Git Hash: 9529cc3
	Board Name: AD7190		Project Name: OPEN_WEIGHT			
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2025-01-12	Revision: + (Unreleased)	
	Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 10	

[3] Project Architecture



	Comments:	Company: OPEN_TRUST_LAB		Variant: PRELIMINARY	Git Hash: 9529cc3
		Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/			Reviewer:	Size: A3	Sheet: 3 of 10

[illegible]

Size:	Sheet:
A4	4 of 10

of 10

Block Description

About

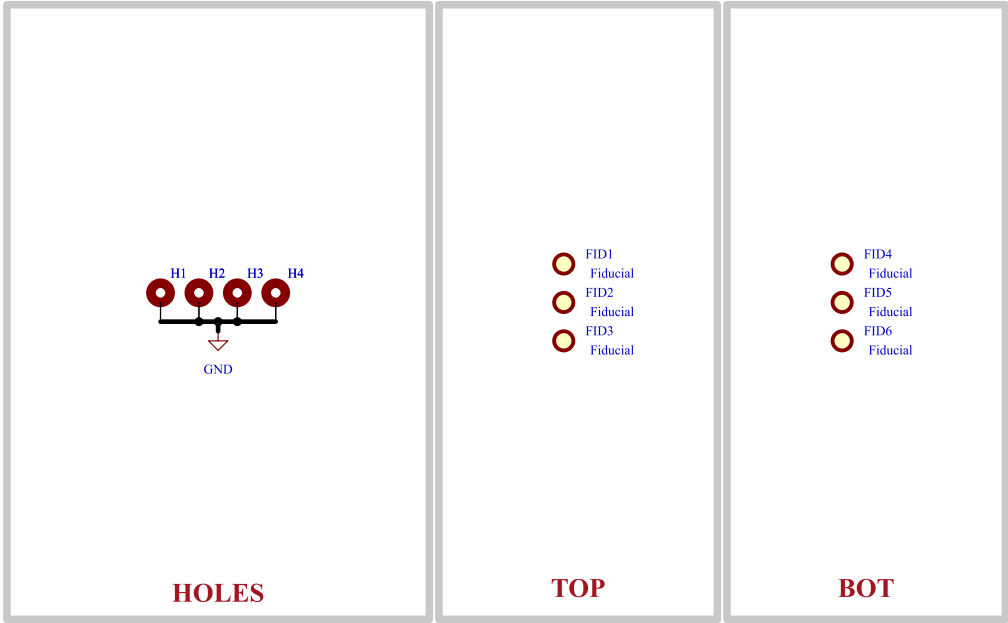
At light-load, the device can operate in Power Save Mode with pulse-frequency-modulation (PFM) to improve the efficiency while keeping a low-output voltage ripple. For very noise-sensitive applications, the device can be forced to PWM Mode operation over the entire load range by pulling the MODE pin high. The external compensation allows optimizing the application for specific conditions. A capacitor connected to the soft-start pin minimizes inrush current at start-up.

Block Description



of **10**

[7]Holes, Fiducials



	Comments: References: Flexible I/O worked examples Flexible I/O source configuration	Company: OPEN_TRUST_LAB		Variant: PRELIMINARY	Git Hash: 9529cc3
	Sheet Title: Sheet Title B	Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Path: /Project Architecture/Holes, Fiducials/	File Name: Section E - Title E.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2025-01-12	Revision: + (Unreleased)
		Reviewer:		Size: A3	Sheet: 7 of 10

[8] Connecteur

DESIGN NOTE: blabla

DESIGN NOTE: blabla

LAYOUT NOTE: blabla

Block Description

Block Description

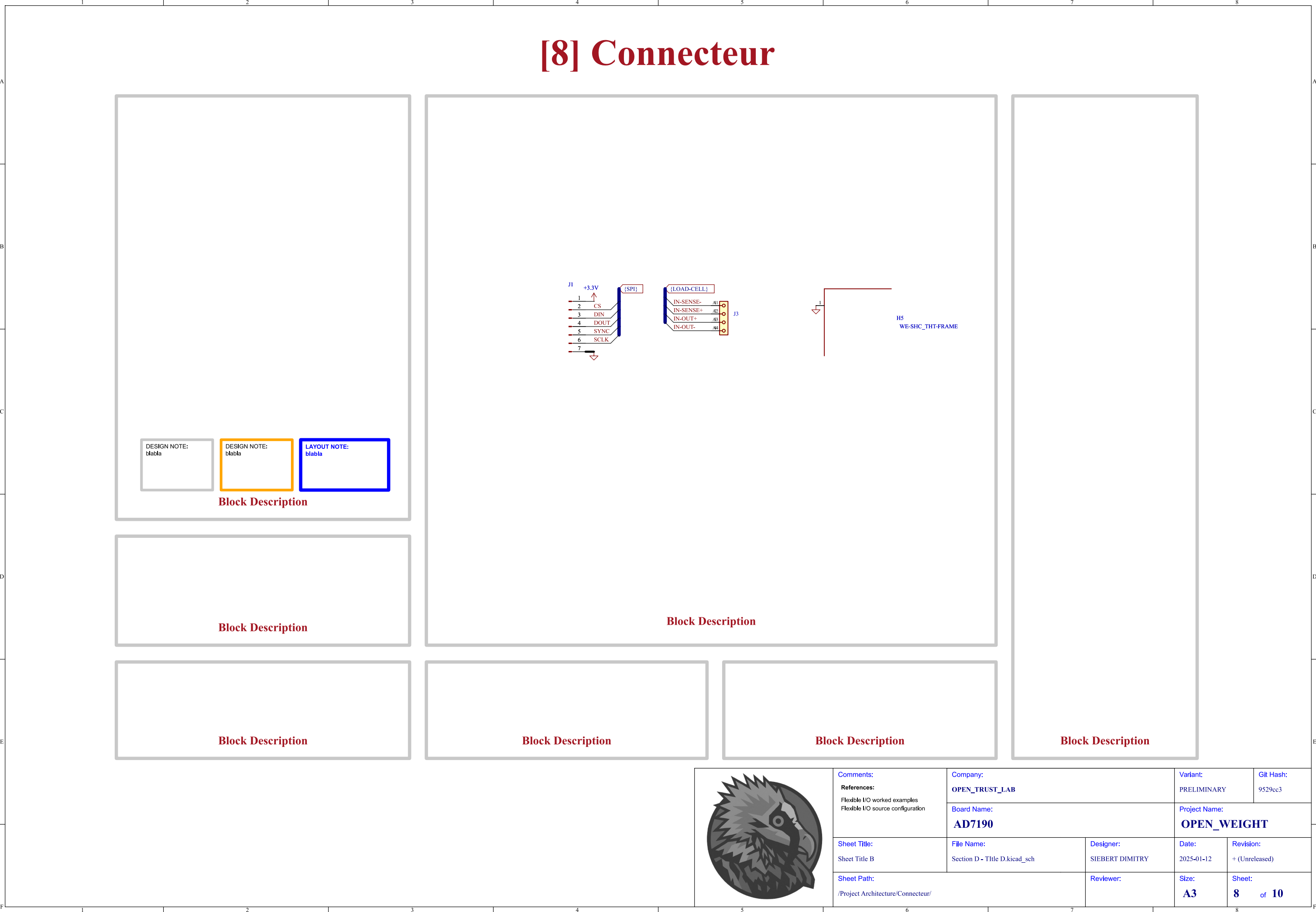
Block Description

Block Description

Block Description

Block Description

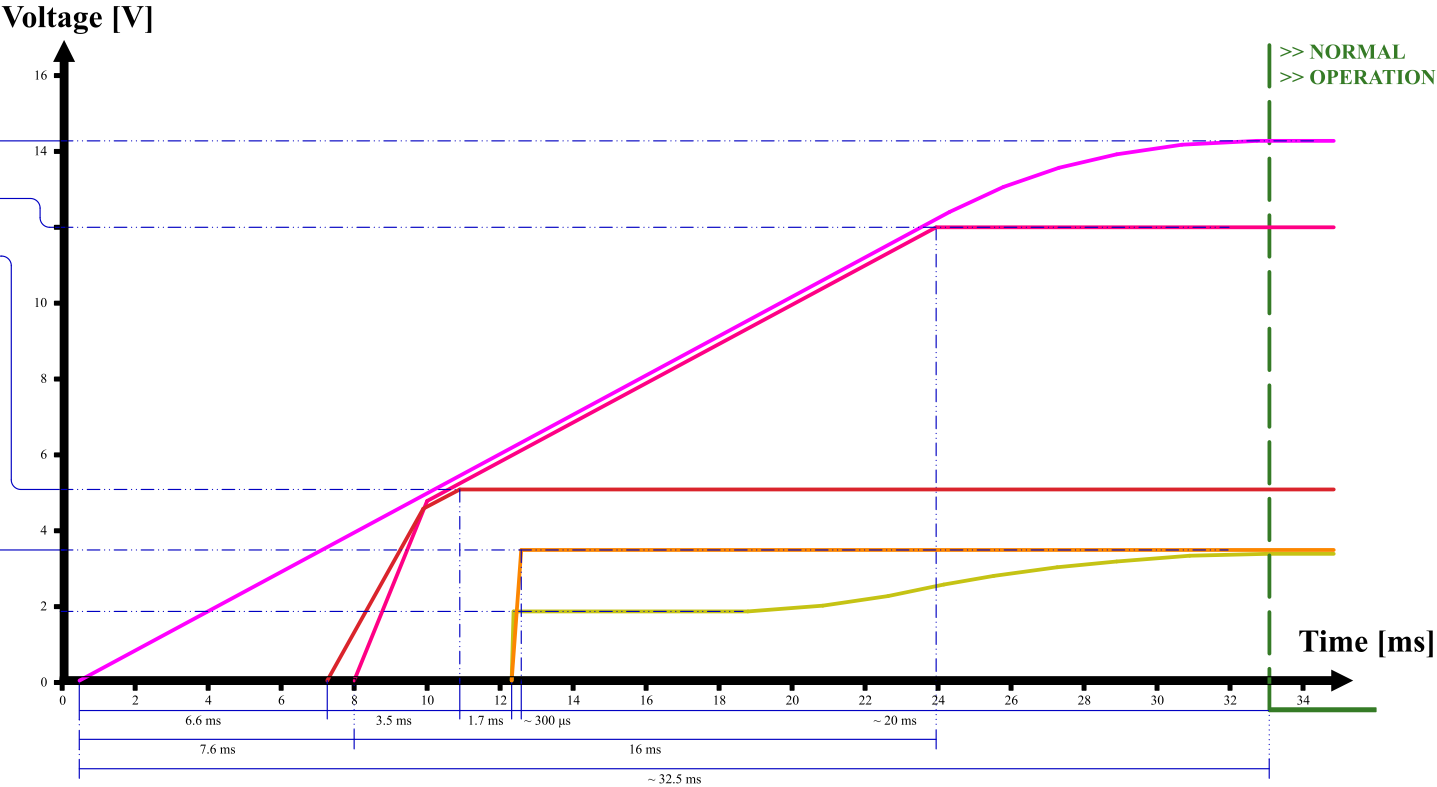
	Comments:	Company: OPEN_TRUST_LAB		Variant: PRELIMINARY	Git Hash: 9529cc3
	References: Flexible I/O worked examples Flexible I/O source configuration	Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Sheet Title B	File Name: Section D - Title D.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Connecteur/		Reviewer:	Size: A3	Sheet: 8 of 10	



9529cc3

[9] Power - Sequencing

NAME	SOURCE	LEVEL
+5VA	LT3042	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



	Comments:	Company: OPEN_TRUST_LAB		Variant: PRELIMINARY	Git Hash: 9529cc3
		Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 9 of 10

[10] Revision History

	Comments:	Company: OPEN_TRUST_LAB		Variant: PRELIMINARY	Git Hash: 9529cc3
		Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 10 of 10